where $\epsilon \in (0, \frac{1}{2})$ is the *effect size* distinguishing P from Q. Let Φ be the Normal cdf, n_{te} the number of samples available for testing, and α the significance level. Then,

Theorem 1. *Given the conditions described in the previous paragraph, the approximate power of the statistic* (2) *is* $\Phi\left(\frac{\epsilon\sqrt{n_{te}} - \Phi^{-1}(1-\alpha)/2}{\sqrt{\frac{1}{4}-\epsilon^2}}\right)$.

See Appendix B for a proof. The power bound in Theorem 1 has an optimal order of magnitude for multi-dimensional problems (Bai & Saranadasa, 1996; Gretton et al., 2012a; Reddi et al., 2015). These are problems with fixed d and $n \to \infty$, where the power bounds do not depend on d.

Remark 1. *We leave for future work the study of quadratic-time C2ST with optimal power in high-dimensional problems (Ramdas et al., 2015). These are problems where the ratio $n/d \to c \in [0, 1]$, and the power bounds depend on d. One possible line of research in this direction is to investigate the power and asymptotic distributions of quadratic-time C2ST statistics $\frac{1}{n_{te}(n_{te}-1)} \sum_{i \neq j} \mathbb{I}[\mathbb{I}(f(z_i, z_j) > \frac{1}{2}) = l_i]$, where the classifier $f(z, z')$ predicts if the examples (z, z') come from the same sample.*

Theorem 1 also illustrates that maximizing the power of a C2ST is a trade-off between two competing objectives: choosing a classifier that *maximizes the test accuracy* ϵ and *maximizing the size of the test set* n_{te}. This relates to the well known bias-variance trade-off in machine learning. Indeed, simple classifiers will miss more nonlinear patterns in the data (leading to smaller test accuracy), but call for less training data (leading to larger test set sizes). On the other hand, flexible classifiers will miss less nonlinear patterns in the data (leading to higher test accuracy), but call for more training data (leading to smaller test sizes). Formally, the relationship between the test accuracy, sample size, and the flexibility of a classifier depends on capacity measures such as the VC-Dimension (Vapnik, 1998). Note that there is no restriction to perform model selection (such as cross-validation) on $\mathcal{D}_{\text{tr}}$.

Remark 2. *We have focused on test statistics* (2) *built on top of the zero-one loss $\ell_{0-1}(y, y') = \mathbb{I}[y = y'] \in \{0, 1\}$. These statistics give rise to Bernoulli random variables, which can exhibit high variance. However, our arguments are readily extended to real-valued binary classification losses. Then, the variance of such real-valued losses would describe the norm of the decision function of the classifier two-sample test, appear in the power expression from Theorem 1, and serve as a hyper-parameter to maximize power as in (Gretton et al., 2012b, Section 3).*[2]

3.3 Interpretability

There are three ways to interpret the result of a C2ST. First, recall that the classifier predictions $f(z_i)$ are estimates of the conditional probabilities $p(l_i = 1|z_i)$ for each of the samples z_i in the test set. Inspecting these probabilities together with the true labels l_i determines which examples were correctly or wrongly labeled by the classifier, with the least or the most confidence. Therefore, the values $f(z_i)$ explain *where* the two distributions differ. Second, C2ST inherit the interpretability of their classifiers to explain which *features* are most important to distinguish distributions, in the same way as the ME test (Jitkrittum et al., 2016). Examples of interpretable features include the filters of the first layer of a neural network, the feature importance of random forests, the weights of a generalized linear model, and so on. Third, C2ST return statistics $\hat{t}$ in interpretable units: these relate to the percentage of samples correctly distinguishable between the two distributions. These interpretable numbers can complement the use of p-values.

3.4 Prior Uses

The reduction of two-sample testing to binary classification was introduced in (Friedman, 2003), studied within the context of information theory in (Pérez-Cruz, 2009; Reid & Williamson, 2011), discussed in (Fukumizu et al., 2009; Gretton et al., 2012a), and analyzed (for the case of linear discriminant analysis) in (Ramdas et al., 2016). The use of binary classifiers for two-sample testing is increasingly common in neuroscience: see (Pereira et al., 2009; Olivetti et al., 2012) and the references therein. Implicitly, binary classifiers also perform two-sample tests in algorithms that discriminate data from noise, such as unsupervised-as-supervised learning (Friedman et al., 2001), noise contrastive estimation (Gutmann & Hyvärinen, 2012), negative sampling (Mikolov et al., 2013), and GANs (Goodfellow et al., 2014).

[2] For a related discussion on this issue, we recommend the insightful comment by Arthur Gretton and Wittawat Jitkrittum, available at `https://openreview.net/forum?id=SJkXfE5xx`.

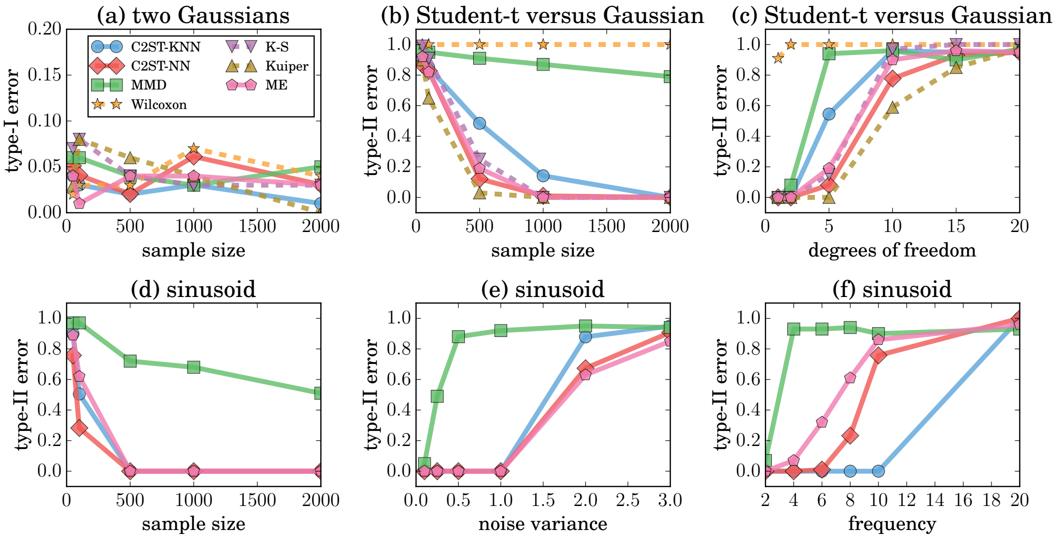

Figure 1: Results (type-I and type-II errors) of our synthetic two-sample test experiments.

4 Experiments on Two-Sample Testing

We study two variants of classifier-based two-sample tests (C2ST): one based on neural networks (C2ST-NN), and one based on k-nearest neighbours (C2ST-KNN). C2ST-NN has one hidden layer of 20 ReLU neurons, and trains for 100 epochs using the Adam optimizer (Kingma & Ba, 2015). C2ST-KNN uses $k = \lfloor n_{\text{tr}}^{1/2} \rfloor$ nearest neighbours for classification. Throughout our experiments, we did not observe a significant improvement in performance when increasing the flexibility of these classifiers (e.g., increasing the number of hidden neurons or decreasing the number of nearest neighbors). When analyzing one-dimensional data, we compare the performance of C2ST-NN and C2ST-KNN against the Wilcoxon-Mann-Whitney test (Wilcoxon, 1945; Mann & Whitney, 1947), the Kolmogorov-Smirnov test (Kolmogorov, 1933; Smirnov, 1939), and the Kuiper test (Kuiper, 1962). In all cases, we also compare the performance of C2ST-NN and C2ST-KNN against the linear-time estimate of the Maximum Mean Discrepancy (MMD) criterion (Gretton et al., 2012a), the ME test (Jitkrittum et al., 2016), and the SCF test (Jitkrittum et al., 2016). We use a significance level $\alpha = 0.05$ across all experiments and tests, unless stated otherwise. We use Gaussian approximations to compute the null distributions of C2ST-NN and C2ST-KNN. We use the implementations of the MMD, ME, and SCF tests gracefully provided by Jitkrittum et al. (2016), the scikit-learn implementation of the Kolmogorov-Smirnov and Wilcoxon tests, and the implementation from `https://github.com/aarchiba/kuiper` of the Kuiper test. The implementation of our experiments is available at `https://github.com/lopezpaz/classifier_tests`.

4.1 Experiments on Two-Sample Testing

Control of type-I errors We start by evaluating the correctness of all the considered two-sample tests by examining if the prescribed significance level $\alpha = 0.05$ upper-bounds their type-I error. To do so, we draw $x_1, \ldots, x_n, y_1, \ldots, y_n \sim \mathcal{N}(0, 1)$, and run each two-sample test on the two samples $\{x_i\}_{i=1}^n$ and $\{y_i\}_{i=1}^n$. In this setup, a type-I error would be to reject the true null hypothesis. Figure 1(a) shows that the type-I error of all tests is upper-bounded by the prescribed significance level, for all $n \in \{25, 50, 100, 500, 1000, 5000, 10000\}$ and 100 random repetitions. Thus, all tests control their type-I error as expected, up to random variations due to finite experiments.

Gaussian versus Student We consider distinguishing between samples drawn from a Normal distribution and samples drawn from a Student's t-distribution with ν degrees of freedom. We shift and scale both samples to exhibit zero-mean and unit-variance. Since the Student's t distribution approaches the Normal distribution as ν increases, a two-sample test must focus on the peaks of the distributions to distinguish one from another. Figure 1(b,c) shows the percentage of type-II errors made by all tests as we vary separately n and ν, over 100 trials (random samples). We set

Problem	n^{te}	ME-full	ME-grid	SCF-full	SCF-grid	MMD-quad	MMD-lin	C2ST-NN
Bayes-Bayes	215	.012	.018	.012	.004	.022	.008	**.002**
Bayes-Deep	216	.954	.034	.688	.180	.906	.262	**1.00**
Bayes-Learn	138	.990	.774	.836	.534	**1.00**	.238	**1.00**
Bayes-Neuro	394	**1.00**	.300	.828	.500	.952	.972	**1.00**
Learn-Deep	149	.956	.052	.656	.138	.876	.500	**1.00**
Learn-Neuro	146	.960	.572	.590	.360	**1.00**	.538	**1.00**

Table 1: Type-I errors (first row) and powers (rest of rows) in distinguishing NIPS papers categories.

Problem	n^{te}	ME-full	ME-grid	SCF-full	SCF-grid	MMD-quad	MMD-lin	C2ST-NN
$\pm$ vs. $\pm$	201	.010	.012	.014	**.002**	.018	.008	**.002**
$+$ vs. $-$	201	.998	.656	**1.00**	.750	**1.00**	.578	.997

Table 2: Type-I errors (first row) and powers (second row) in distinguishing facial expressions.

$n = 2000$ when ν varies, and let $\nu = 3$ when n varies. The Wilcoxon-Mann-Whitney exhibits the worst performance, as expected (since the ranks mean of the Gaussian and Student's t distributions coincide) in this experiment. The best performing method is the the one-dimensional Kuiper test, followed closely by the multi-dimensional tests C2ST-NN and ME.

Independence testing on sinusoids For completeness, we showcase the use two-sample tests to measure statistical dependence. This can be done, as described in Section 2, by performing a two-sample test between the observed data $\{(x_i, y_i)\}_{i=1}^n$ and $\{(x_i, y_{\sigma(i)})\}_{i=1}^n$, where σ is a random permutation. Since the distributions $P(X)P(Y)$ and $P(X,Y)$ are bivariate, only the C2ST-NN, C2ST-KNN, MMD, and ME tests compete in this task. We draw (x_i, y_i) according to the generative model $x_i \sim \mathcal{N}(0,1)$, $\epsilon_i \sim \mathcal{N}(0,\gamma^2)$, and $y_i \sim \cos(\delta x_i) + \epsilon_i$. Here, x_i are iid examples from the random variable X, and y_i are iid examples from the random variable Y. Thus, the statistical dependence between X and Y weakens as we increase the frequency δ of the sinusoid, or increase the variance γ^2 of the additive noise. Figure 1(d,e,f) shows the percentage of type-II errors made by C2ST-NN, C2ST-KNN, MMD, and ME as we vary separately n, δ, and γ over 100 trials. We let $n = 2000$, $\delta = 1$, $\gamma = 0.25$ when fixed. Figure 1(d,e,f) reveals that among all tests, C2ST-NN is the most efficient in terms of sample size, C2ST-KNN is the most robust with respect to high-frequency variations, and that C2ST-NN and ME are the most robust with respect to additive noise.

Distinguishing between NIPS articles We consider the problem of distinguishing between some of the categories of the 5903 articles published in the Neural Information Processing Systems (NIPS) conference from 1988 to 2015, as discussed in Jitkrittum et al. (2016). We consider articles on Bayesian inference (Bayes), neuroscience (Neuro), deep learning (Deep), and statistical learning theory (Learn). Table 1 shows the type-I errors (Bayes-Bayes row) and powers (rest of rows) for the tests reported in (Jitkrittum et al., 2016), together with C2ST-NN, at a significance level $\alpha = 0.01$, when averaged over 500 trials. In these experiments, C2ST-NN achieves maximum power, while upper-bounding its type-I error by α.

Distinguishing between facial expressions Finally, we apply C2ST-NN to the problem of distinguishing between positive (happy, neutral, surprised) and negative (afraid, angry, disgusted) facial expressions from the Karolinska Directed Emotional Faces dataset, as discussed in (Jitkrittum et al., 2016). See the fourth plot of Figure 2, first two-rows, for one example of each of these six emotions. Table 2 shows the type-I errors ($\pm$ vs $\pm$ row) and the powers ($+$ vs $-$ row) for the tests reported in (Jitkrittum et al., 2016), together with C2ST-NN, at $\alpha = 0.01$, averaged over 500 trials. C2ST-NN achieves a near-optimal power, only marginally behind the perfect results of SCF-full and MMD-quad.

5 Experiments on Generative Adversarial Network Evaluation

Since effective generative models will produce examples barely distinguishable from real data, two-sample tests arise as a natural alternative to evaluate generative models. Particularly, our interest is to evaluate the sample quality of generative models with intractable likelihoods, such as GANs (Goodfellow et al., 2014). GANs implement the adversarial game

$$\min_g \max_d \mathop{\mathbb{E}}_{x\sim P(X)} \left[\log(d(x))\right] + \mathop{\mathbb{E}}_{z\sim P(Z)} \left[\log(1-d(g(z)))\right], \tag{3}$$